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Mg Anode Corrosion in Aqueous Electrolytes and Implications for Mg-Air Batteries

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Aqueous Mg-air primary batteries possess many favorable attributes for energy storage because Mg is affordable, abundant, and lightweight. However, parasitic corrosion of Mg in aqueous electrolytes generates H_2 and surprisingly increases with increasing current density during battery discharge (Mg oxidation), limiting the faradaic efficiency of aqueous Mg batteries. In this study, differential electrochemical mass spectrometry and H_2 pressure rise measurements were used to characterize Mg corrosion in Mg-air batteries employing aqueous electrolytes with salts (NaCl, $NaNO_3$, $NaPO_4$, and a NaCl/$NaPO_4$ mixture) that provide various degrees of Mg passivation. H_2 evolution rates were highest in NaCl electrolytes and lowest in $NaNO_3$ electrolytes. However, $NaNO_3$ salts reduced the H_2-evolving corrosion rate at the expense of introducing a nitrate to nitrite corrosion reaction into the battery. The combined Mg corrosion rate in the nitrate-based electrolyte was still lowest among those electrolytes studied. The nitrate to nitrite corrosion reaction also lowered the magnitude of the Mg anodic potential and therefore decreased the overall Mg-O_2 battery voltage compared to the NaCl electrolyte. Nevertheless, Mg-O_2 batteries utilizing a $NaNO_3$ electrolyte allowed for 60% larger discharge capacity and 50% higher Mg oxidation faradaic efficiency compared to a NaCl electrolyte.
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Increased demand for electrical energy storage has sparked a renewed interest into battery chemistries with theoretical energy densities that are much larger than lithium ion batteries. Among these chemistries, a tremendous amount of recent work has been devoted to metal-air batteries such as Li-O_2,[1,2] Na-O_2,[3] and Zn-O_2.[4] However, much less attention has been paid to Al or Mg-O_2 batteries even though Mg and Al are lightweight, abundant, and environmentally friendly.[5] Although these batteries are not electrically rechargeable, there is commercial interest in them as standby power sources and as range extenders in electric vehicles because of their inherent safety and long standby lifetimes. Mechanically rechargeable aqueous Mg-O_2 batteries, those in which a fully oxidized Mg electrode is replaced with a fresh electrode, are especially attractive since they can function with saltwater electrolytes and have high theoretical voltages. Therefore, they can be used effectively as underwater batteries using dissolved O_2 in seawater. In addition, they could function effectively as standby range extenders for traditional Li ion battery packs since the Mg metal has no self-discharge until placed in the electrolyte. The desirable reactions for the aqueous Mg-O_2 battery are shown below:

$$\text{Anode: } Mg \rightarrow Mg^{2+} + 2e^- \quad [1]$$

$$\text{Cathode: } O_2 + 2H_2O + 4e^- \rightarrow 4OH^- \quad [2]$$

$$\text{Overall: } 2Mg + O_2 + 2H_2O \rightarrow 2Mg(OH)_2 \quad [3]$$

Unfortunately, Mg is a reactive metal that corrodes significantly in the presence of aqueous electrolytes, decreasing the faradaic efficiency of the battery.[6] The parasitic corrosion reactions that occur on the Mg anode are shown below:

$$Oxidation: Mg \rightarrow Mg^{2+} + 2e^- \quad [4]$$

$$Reduction: 2H_2O + 2e^- \rightarrow H_2 + 2OH^- \quad [5]$$

$$Overall: Mg + 2H_2O \rightarrow H_2 + Mg(OH)_2 \quad [6]$$

Numerous studies have focused on understanding the fundamental corrosion reaction of Mg in aqueous electrolytes,[7–9] and in particular, understanding why the corrosion rate actually increases on the Mg anode during oxidation, a phenomenon referred to as the negative difference effect (NDE).[10,11] This effect is particularly unfortunate for the Mg-O_2 battery since it means that the corrosion rate increases during battery discharge. In order to reduce the corrosion rate, most of the literature focuses on alloying the Mg anode,[12,13] although adding corrosion inhibitors that passivate the Mg surface with surface films has also been investigated.[14,15] Unfortunately, the passivating surface film is typically Mg^{2+} impermeable, which obviously inhibits Mg dissolution (Equation 1) necessary for Mg-O_2 battery operation. Therefore, in a practical battery, a compromise between inhibiting corrosion and allowing Mg dissolution is required. Only a few papers have reported experimental results on operational performance of aqueous Mg-O_2 batteries. Among these papers, the effect of nanostructured Mg, porous Mg, and alloyed Mg anodes has been investigated.[16–18] In the work reported here, differential electrochemical mass spectroscopy (DEMS) and pressure rise measurements were used to study operating aqueous Mg-O_2 cells and fundamentally understand the role of the electrolyte salt on Mg corrosion. The electrolytes studied here included chloride-based electrolytes, which are known to result in substantial Mg corrosion,[8,9,19] phosphate and nitrate-based electrolytes, which are known to suppress Mg corrosion,[16,20,21] and mixtures of chloride and phosphate salts, which we hypothesized may allow suppressed susceptibility to corrosion, yet afford facile Mg oxidation during anodic polarization.

Experimental

Materials characterization.—NaCl (99.5%), HNa_2PO_4 (99.9%), and $NaNO_3$ (99%) salts were purchased from Fisher Scientific and used as received. Electrolyte solutions were made by mixing these salts with type 1 ultrapure water at a concentration of either 0.1 mol/L or 0.5 mol/L. Potassium iodide (99.5%), sulfuric acid, standardized hydrochloric acid (0.01 N), standardized sodium thiosulfate (0.1 N), 1 wt% starch in H_2O, and ammonium molybdate (99.9%) were all purchased form Sigma Aldrich. Commercial MnO_2 cathodes (E-4) were purchased from Electric Fuel Ltd and used as received in the full battery cell configuration. High purity planar Mg sheets (250 μm thick) were purchased from Gallium Source and used in all experiments. From the manufacturers specifications, the impurities in the Mg are

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as follows: 30 ppm Al, 50 ppm Mn, 50 ppm Si, 40 ppm Ca, 30 ppm Fe, 2 ppm Ni, 5 ppm Pb, 50 ppm Zn. Mg was used without any pretreatment to maintain consistency between experiments. Polishing the surface of Mg alloys has been shown to decrease the initial corrosion rate due to removal of contaminants on the alloy surface.[13] However, we obtained significantly different corrosion rates from one polished sample to the next. Therefore, to maintain experiment-to-experiment consistency, we chose to use the Mg as-received (unpolished). Nafion 117 membranes with a 28 mm diameter were used in the H-cell to separate the anode chamber from the cathode chamber. The coiled Pt wire (23 cm) counter electrode and a Ag/AgCl reference electrode with glass frit were purchased from BASi Inc. The glass H-cell and the glass bulk electrolysis cell (4 port, ~30 ml volume) were custom designed and constructed by Adams & Chittenden Scientific Glass (Berkeley, CA). X-ray diffraction (XRD) measurements of the Mg were performed on a Bruker D2 Phaser diffractometer using CuKα radiation. Diffraction patterns were scanned over an angular range of 10–90°(2θ).

Electrochemical measurements.—Electrochemical measurements were performed on a Biologic SP-300 potentiostat. When using the H-cell and bulk electrolysis cell, the voltages of both the counter electrode (Pt wire) and working electrode (Mg foil, ~1 cm^2 exposed to electrolyte) were recorded simultaneously vs. a Ag/AgCl reference electrode in the Mg anode portion of the H-cell. All potentials are reported versus Ag/AgCl, except for full cell battery experiments. Impedance and linear sweep voltammetry (LSV) experiments were performed in air in the bulk electrolysis cell with vigorous stirring using a magnetic stir bar. Impedance experiments were performed with an amplitude of 10 mV over a frequency range of 0.1–2 × 10^6 Hz. LSV experiments were performed by sweeping the Mg working electrode potential from open circuit potential (OCV) at a scan rate of 5 mV/s.

Differential electrochemical mass spectroscopy (DEMS) and gas pressure rise measurements.—DEMS and pressure rise measurements were performed using a system similar to the one previously published by McCloskey et al.[22] Battery cells were constructed in a 2-electrode cell similar to the configuration previously published.[23] The 2-electrode battery cell consisted of a 0.4375" diameter Mg disk (250 μm thick) anode, a 0.5" diameter filter paper separator (QMA grade, Whatman), and a cathode composed of commercial MnO_2 E-4 material (12 mm diameter, 0.472"). The electrolyte for the battery cell was 100 μl of 0.5 M NaCl or $NaNO_3$ in DI water. All components of the battery cell were assembled in a glove box. Gases evolved from the battery cell were analyzed in the headspace above the cathode in a similar manner to those previously published.[24]

The H-cell configuration is shown in Figure 1. The gases in each side of the cell can be analyzed individually using the mass spectrometer (Stanford Research Systems, RGA200) and an in line pressure transducer (Omega Engineering Inc., PX419). An Al electrode holder was used to hold the Mg foil WE electrode so that the Mg foil is immersed in the electrolyte, but no portion of the Al holder is immersed. The Mg WE chamber electrolyte was stirred vigorously with a stir bar over the course of the experiment to limit mass transport effects. The volume of each chamber of the H-cell was calibrated using a volume expansion technique and the ideal gas law. The volumes of the 3-port chamber (Mg foil and Ag/AgCl reference electrode) and the 2-port chamber (Pt wire counter electrode) of the H-cell were 11.1±0.2 ml and 9.4±0.2 ml, respectively. The uncertainty in the volume arises from the Nafion membrane deflecting when the pressure difference between each chamber of the H-cell increases. The volume of the capillaries between the H-cell and the pressure transducer are 1.2 ml. The 3-port and 2-port chamber of the H-cell are each filled with 5 ml of electrolyte during each experiment so that the total headspace of gas above the electrolyte in the 3-port chamber is calculated to be 7.3 ± 0.2 ml. Initially, the headspace above the electrolyte was filled with Ar. As H_2 was evolved from the Mg foil submersed in the electrolyte, the pressure transducer measures changes in pressure of the 3-port chamber containing the Mg foil so that the moles of H_2 evolved could be quantified directly. Additionally, the gases evolved from Mg could be quantified by pulsing small volumes (500 μL) of gas above the 3-port headspace to a mass spectrometer. Only H_2, and no other gas, was found to evolve during any of the measurements reported in this article (see Figure S1 for mass spectrometry gas analysis during anodic and cathodic chronopotentiometry at a Mg electrode). Furthermore, H_2 evolution was identical under Ar and O_2 headspaces (Figure S2), and therefore electrochemical measurements were performed under an Ar headspace for safety purposes unless otherwise noted.

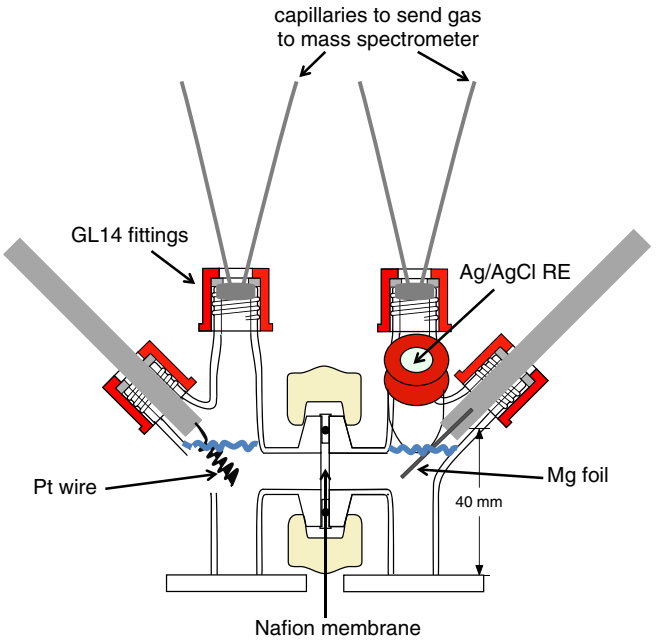


Figure 1. Electrochemical H-cell designed to measure gas evolution/consumption from the Mg foil working electrode (WE) and Pt wire counter electrode (CE) individually. The WE and CE chambers are separated by a Nafion 117 membrane. Each chamber is filled with 5 ml electrolyte and the Mg WE chamber is stirred to limit mass transport effects. Image adapted from an illustration provided by Tom Adams, Adams & Chittenden Scientific Glassware.

NO_2^- Titration.—Titration of the $NaNO_3$ electrolyte to detect NO_2^- was performed in a manner similar to the peroxide titration previously developed by McCloskey et al.[25] Briefly, the following three reagents were added, in succession, to an aliquot of electrolyte extracted from the Mg electrode H-cell chamber: 1 mL of 2 wt% KI in H_2O, 1 mL of 3.5 M H_2SO_4, and 50 μL of a molybdate-based catalyst solution. The catalyst solution was prepared by dissolving 1 g ammonium molybdate in 10 mL of 6 N ammonium hydroxide, followed by adding 3 g ammonium nitrate and diluting the solution to 50 mL total using ultrapure H_2O. The extracted electrolyte solution turns yellow upon reagent addition due to I_2 formation, and the I_2 is immediately titrated to a faint straw color using 0.01 N NaS_2O_3. At this point, ~0.5 mL of a 1% starch indicator is added to the solution, which turns dark blue, and the titration is continued until the solution turns clear. The total time necessary for the peroxide titration after the reagents are added is typically ~1 minute. This titration protocol, originally developed by McCloskey et al. for peroxide quantification in Li-O_2 batteries, can be extended to quantify NO_2^-, which acts in place of peroxide as the I^- oxidizer, as long as it is performed in an inert atmosphere (NO_2^- spontaneously converts to NO in air). The NO_2^- titration accuracy was confirmed by making a solution with a known concentration of $LiNO_2$ (5.5 mg/ml) in water (200 μL) and then titrating using the previously described peroxide titration.[25] A titration accuracy of >90% was achieved in the control experiment. Titration of the $NaNO_3$ electrolyte in the H-cell was performed by

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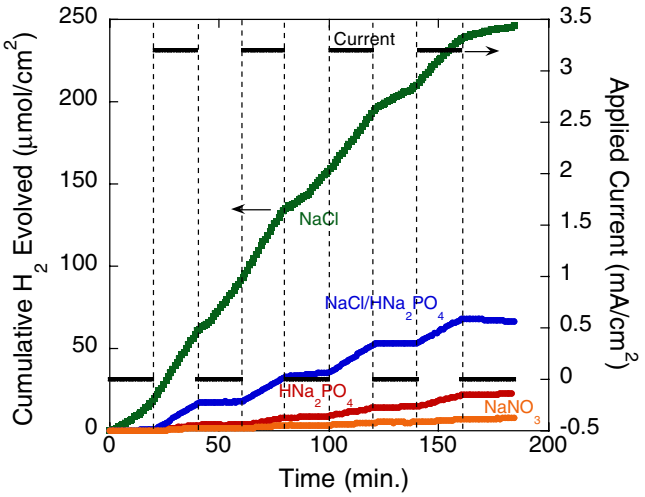


Figure 2. H_2 evolution from a Mg working electrode (1.2 cm^2 surface area) in 0.5 M NaCl (green), 0.5 M NaCl / 0.05 M HNa_2PO_4 (blue), 0.5 M HNa_2PO_4 (red), and 0.5 M $NaNO_3$ (orange) electrolytes. Four consecutive anodic chronopotentiometric scans (20 min at 3.2 mA/cm^2) are shown in between holding the electrode at OCV for 20 minutes.

removing 150 µL aliquots of $NaNO_3$ electrolyte from the Mg chamber of the H-cell with a syringe. Aliquots were removed and titrated at 15 minute intervals during both OCV and discharge of the Mg anode. The $NaNO_3$ aliquots were quickly transferred from the syringe into an Ar filled vial (with septum cap) before titration under Ar. The quantity of NO_2^- found in the 150 µL aliquot was scaled to the amount that would be found in the whole electrolyte volume assuming the solution was well mixed.

Results

Figure 1 shows a schematic of the electrochemical H-cell that was developed to isolate the Mg electrode from the Pt electrode so that gases evolved/consumed at each electrode could be analyzed individually without influencing one another. The Pt wire counter electrode (CE) is located in the 2-port chamber of the H-cell and the Ag/AgCl reference electrode (RE) and Mg foil working electrode (WE) are located in the 3-port chamber. For the purposes of this study, we focused on the electrochemistry and gas evolution in the Mg WE chamber in order to understand the fundamental Mg corrosion reaction in various aqueous electrolytes. Differential electrochemical mass spectroscopy (DEMS) of the gases evolved in the Mg WE chamber confirmed that H_2 was the only gas evolved at OCV both during reduction and oxidation at the Mg WE (Figure S1).

The Mg corrosion rate is strongly influenced by the electrolyte employed. Figure 2 shows the cumulative H_2 evolved (normalized to the surface area of Mg foil exposed to electrolyte) from the Mg WE chamber of the H-cell in four aqueous electrolytes, selected given their use in previous reports on Mg-O_2 batteries and Mg corrosion studies,[16,20,21] during anodic chronopotentiometry (Mg oxidation). A fifth aqueous electrolyte, 0.5 M Na_2SO_4, was also characterized (Figure S3), but was found to result in substantially higher H_2 evolution than phosphate-based electrolytes and was therefore not further explored. Four consecutive 20 min., 3.2 mA cm^{-2} anodic steps, in which each step is followed by a 20 min OCV, were performed. In a typical experiment, the initial pH of the NaCl and $NaNO_3$ electrolytes was 5 and increased to ~8 during the course of data collection. The HNa_2PO_4 and NaCl/HNa_2PO_4 mixed electrolytes initially started at pH 8 and increased to pH 9. For 0.5 M HNa_2PO_4, 0.5 M NaCl / 0.05 M HNa_2PO_4 mixture, and 0.5 M $NaNO_3$ electrolytes, negligible H_2 evolution (Mg corrosion) was observed at OCV when compared to the pure 0.5 M NaCl electrolyte, suggesting that these electrolyte salts passivate the Mg surface more effectively than NaCl. For all four electrolytes, the H_2 evolution increases once the galvanostatic oxidation of Mg begins (i.e. Mg anodic dissolution) and returns to the OCV H_2 evolution rate (i.e. inherent Mg corrosion) at the end of the galvanostatic oxidation. The seemingly thermodynamic-defying increase of H_2 evolution during anodic Mg dissolution, termed the negative difference effect (NDE), is consistent with previous reports[7,10] and is observed to occur to varying extents in the various electrolytes. Consistent with previous reports,[16,20,21] the $NaNO_3$ electrolyte appears to almost completely inhibit corrosion, both at OCV and during oxidation, given the extremely small H_2 evolution observed. However, we will show later that the suppressed H_2 evolution is at the expense of the introduction of a second corrosion reaction.

The Mg voltage, H_2 evolution rate (r_i), and faradaic efficiency of productive Mg^{2+} dissolution (i.e. that not due to the corrosion reaction) in each electrolyte are shown in Table I. The OCV in NaCl is close to that expected for the standard potential of Mg.[26] Mg^{2+} faradaic efficiency is the most important metric for the Mg anode in an Mg-air battery during discharge. Mg^{2+} faradaic efficiency was calculated by converting the H_2 evolution rate to a faradaic current as shown in the footnote in Table I, where F is Faraday's constant and z is the valency (value of +2 for Mg). Figure 1 and Table I show that NaCl and the mixture of NaCl and HNa_2PO_4 have the lowest Mg^{2+} faradaic efficiency due to large H_2 corrosion rates during discharge. The addition of a small amount of HNa_2PO_4 buffer to a traditional NaCl electrolyte has been shown to be an effective method to decrease the OCV H_2 evolution and Mg corrosion, without affecting the discharge voltage significantly.[14,27] The mixture of HNa_2PO_4 and NaCl does indeed exhibit lower OCV H_2 evolution corrosion, however, the H_2 evolution during Mg oxidation resulting from the negative difference effect does not decrease in the presence of small amounts of HNa_2PO_4. From Table I, pure 0.5 M HNa_2PO_4 electrolyte increased the Mg^{2+} faradaic efficiency compared to NaCl electrolytes, but it also simultaneously introduced extraordinarily large overpotentials during discharge (electrode potential of +2.3 V vs Ag/AgCl during oxidation at 3.2 mA/cm^2, whereas OCV was −1.52 vs. Ag/AgCl, see Figure S4 and S5). The very large overpotential during discharge

Table I. Comparison of Mg Anode Properties in Different Electrolytes.

Electrolyte	OCV (V)	Discharge* Potential (V)	OCV H_2 Evolution (r_{OCV}, µmol $min.^{-1}cm^{-2}$)	Discharge* H_2 Evolution ($r_{discharge}$, µmol $min.^{-1}cm^{-2}$)	Mg^{2+} Faradaic efficiency (%)[a]
0.5 M NaCl	−1.61	−1.58	0.79	1.39	40
0.5 M NaCl/0.05 M HNa_2PO_4	−1.58	−1.54	0.04	0.73	31
0.5 M HNa_2PO_4	−1.52	2.3	0.01	0.35	66
0.5 M $NaNO_3$	−1.4	−1.35	0.01	0.08	93**

[a]Mg^{2+} faradaic efficiency defined as: $\frac{(i_{applied}-i_{H2})}{i_{applied}} \times 100; i_{H2} = (r_{discharge} - r_{OCV})Fz, i_{applied} = 3.2\frac{mA}{cm^2}$.

*3.2 mA/cm^2 discharge.

**Based on H_2 corrosion reaction only.

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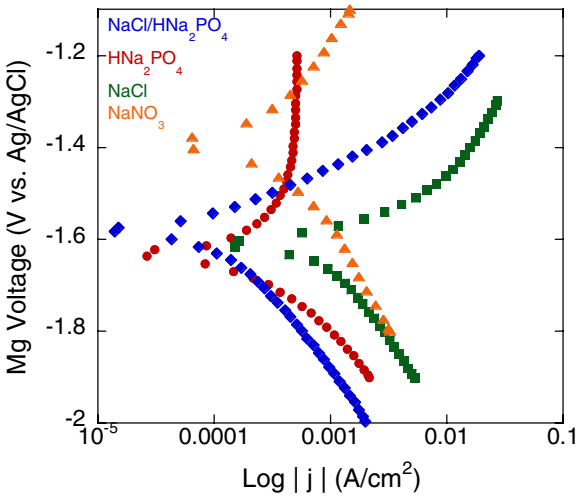


Figure 3. (a) Apparent Tafel plots in the bulk electrolysis cell for 0.5 M NaCl (green data), 0.5 M NaCl / 0.05 M HNa_2PO_4 (blue data), 0.5 M HNa_2PO_4 (red data), and 0.5 M $NaNO_3$ (orange data) electrolytes with a Mg foil working electrode, Pt wire counter electrode, and Ag/AgCl reference electrode.

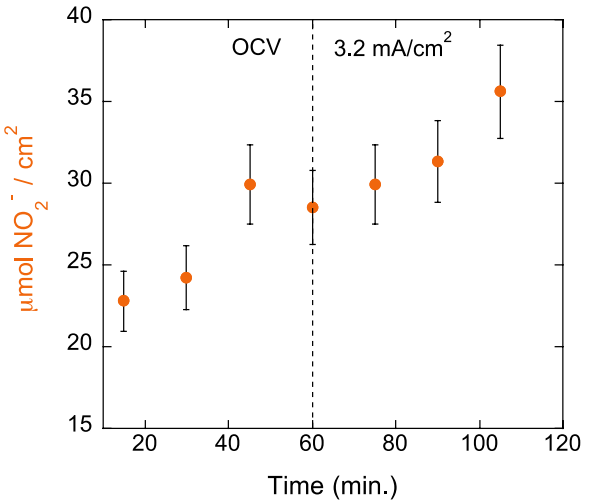


Figure 5. Nitrite titration results displaying the quantity of nitrite (NO_2^-) formed in the Mg chamber of the H-cell in 0.5 M $NaNO_3$ electrolyte during OCV and 3.2 mA/cm^2 discharge.

indicates the pure HNa_2PO_4 electrolyte passivates the Mg surface too strongly, presumably by introducing a very resistive layer between the metal and the electrolyte interface. The $NaNO_3$ electrolyte displays the smallest H_2 evolution and the highest apparent faradaic efficiency for Mg^{2+} dissolution. Unfortunately, the Mg OCV and discharge voltage are reduced in the $NaNO_3$ electrolyte, hence the overall battery cell voltage would be decreased compared to the NaCl and HNa_2PO_4 electrolytes.

If only considering the gas evolution results (Figure 1 and Table I), the corrosion protection provided by the pure 0.5 M $NaNO_3$ electrolyte without any NaCl seems to be the most favorable electrolyte for an aqueous Mg-O_2 battery since it evolves the smallest quantity of hydrogen (least corrosion) and hence displays the highest apparent Mg^{2+} faradaic efficiency.[16] However, the electrolyte salt choice also affects the electrochemistry of the battery, which can be seen more clearly by examining the apparent Tafel plots and Nyquist plots for Mg in each electrolyte, as shown in Figures 3 and 4, respectively.

Figure 3 shows apparent (not iR corrected) Tafel plots for the Mg foil working electrode in the same electrolytes as in Figure 1. The voltage was scanned cathodically (Mg or H^+ reduction) and anodically (Mg oxidation) at a rate of 5 mV/s. The Tafel slope for anodic dissolution of Mg in NaCl electrolyte remains small even at modest current densities, a desirable feature of Mg anodes in Mg-O_2 batteries. However, the Mg voltage quickly increased as the anodic current increased in the HNa_2PO_4 electrolyte, yielding a large (almost vertical) Tafel slope and indicating the difficulty of Mg^{2+} dissolution through the passivating layer on the Mg surface at larger current densities, in agreement with the results in Table I for HNa_2PO_4. Interestingly, mass spectroscopy measurements during Mg oxidation in the 0.5 M HNa_2PO_4 electrolyte revealed H_2 evolution from the Mg occurred even at potentials as large as 2.2 V (Figure S4 and S5). This result implies that the potential at the Mg surface and the electrolyte surface are significantly different in 0.5 M HNa_2PO_4 due to the presence of a high impedance passivating film (which is confirmed in Figure 4, as will be discussed later).

Figure 3 clearly shows the higher OCV of the Mg electrode in the $NaNO_3$ electrolyte compared to the NaCl and HNa_2PO_4 electrolytes. The higher Mg potential in the $NaNO_3$ electrolyte is unfavorable for a Mg-O_2 battery, as it results in a lower discharge voltage. Additionally, for the $NaNO_3$ electrolyte during Mg anodic dissolution, an intermediate Tafel slope is evident compared to the HNa_2PO_4 and NaCl electrolytes. $NaNO_3$ inhibits OCV H_2 evolution, and still appears to minimize the H_2 evolution at moderate current densities. However, the different OCV and H_2 evolution for the $NaNO_3$ electrolyte suggest

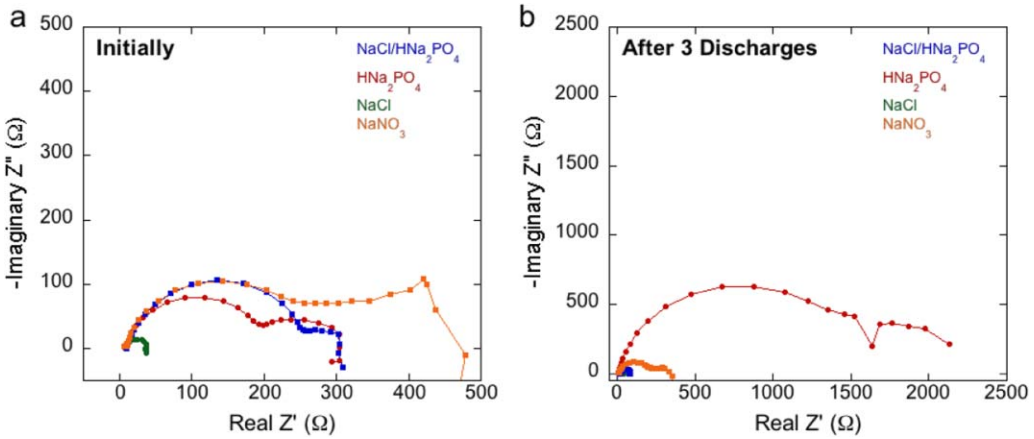


Figure 4. Nyquist plots (a) prior to discharges and (b) after 3 20 min., 3.2 mA/cm^2 discharges in the bulk electrolysis cell for 0.5 M NaCl (green data), 0.5 M NaCl / 0.05 M HNa_2PO_4 (blue data), 0.5 M HNa_2PO_4 (red data), and 0.5 M $NaNO_3$ (orange data) electrolytes with a Mg foil working electrode (~1cm^2), Pt wire counter electrode, and Ag/AgCl reference electrode.

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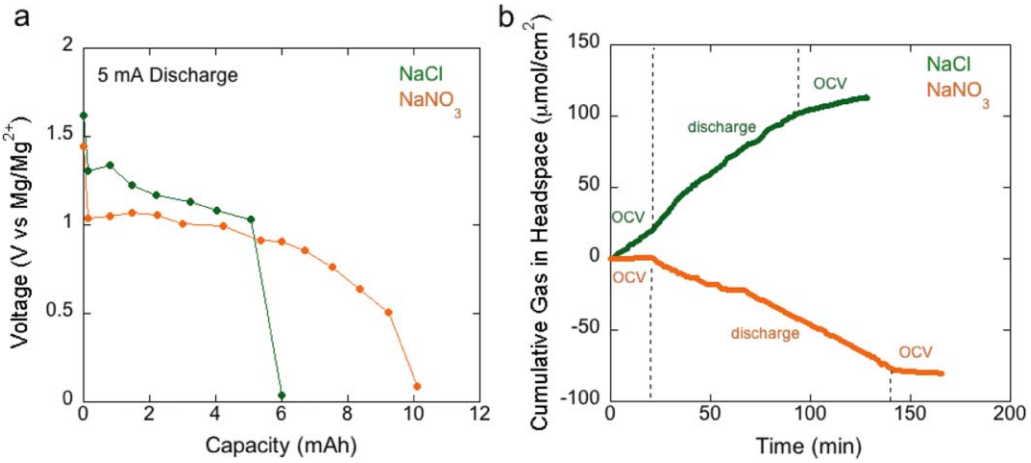


Figure 6. Full Mg-O_2 battery cell (a) voltage and (b) gas evolution/consumption during a 5 mA/cm^2 discharge in 0.5 M NaCl (green data) and 0.5 M $NaNO_3$ (orange data).

that a different corrosion reaction than Eq. 5 may be occurring in this electrolyte, as will be shown later in Figure 5.

A trade-off exists between the ability of an electrolyte to passivate the Mg surface (decrease corrosion and H_2 evolution) and allow Mg^{2+} dissolution through the passivating layer during discharge. The effect of the passivating layer was examined by inspecting the Nyquist plots at OCV for each electrolyte as shown in Figures 4a, 4b. Figure 4a shows the Nyquist plot for the Mg working electrode during OCV prior to discharges in the same electrolytes as in the previous figures. The ability of the electrolyte to form a passivating layer on the Mg surface is directly related to the resistance (intercept on the x-axis) from the Nyquist plot. In the Mg corrosion literature, the high frequency semicircle is ascribed to the charge transfer resistance at the Mg surface and the low frequency semicircle to the mass transfer resistance (diffusion of Mg^{2+}) through the solid film (corrosion product) on the Mg.[27,28] A weak passivating film existed on the Mg surface in NaCl, leading to facile Mg^{2+} dissolution at large current densities and simultaneously large Mg corrosion rates. In contrast, the pure HNa_2PO_4 and $NaNO_3$ electrolyte exhibited clear evidence of a significant mass transfer resistance, consistent with the higher slopes for anodic Mg^{2+} dissolution observed in the apparent Tafel plots in Figure 3.[29] The NaCl electrolyte displayed the smallest resistance (highest corrosion) and the $NaNO_3$ electrolyte displayed the highest resistance (lowest corrosion).

Figure 4b shows the Nyquist plot for Mg in each electrolyte after 3 consecutive 20 min. discharges (3.2 mA/cm^2). The resistance in the $NaNO_3$ electrolyte did not change significantly after discharging, hence the passivating layer seems to be unaffected by discharge. Interestingly, in the HNa_2PO_4 electrolyte, the resistance increased significantly after discharge, indicating that a stronger passivating film is formed on Mg during discharge. Post-mortem XRD on electrodes indicate substantial $Mg(OH)_2$ surface coverage occurs during corrosion in NaCl electrolytes (see supporting information Figure S6). Although no surface species were observed in the phosphate-based electrolytes using XRD, a thin, highly compact, undetectable surface layer likely formed given the large impedance rise.

Figures 3 and 4 indicate the importance of the role of the passivating film formed on the Mg surface in limiting OCV Mg corrosion and also in allowing Mg^{2+} dissolution during Mg oxidation (battery discharge).[19,30] As mentioned previously, the $NaNO_3$ electrolyte seems to inhibit Mg corrosion, while simultaneously allowing Mg^{2+} dissolution during battery discharge. Unfortunately, as indicated previously, the OCV and limited H_2 evolution behavior suggest that another corrosion reaction may be occurring. In fact, the nitrate (NO_3^-) to nitrite (NO_2^-) reaction is known to occur at low overpotentials in aqueous electrolytes as shown in Equation 7.

$$NO_3^- + H_2O + 2e^- \rightarrow NO_2^- + 2OH^- \quad (U^0 = 0.01(V \text{ vs. } SHE)) \quad [7]$$

In order to test the hypothesis that Eqn. 7 is the dominant cathodic corrosion reaction (instead of H_2 evolution) on Mg in the $NaNO_3$ electrolyte, a nitrite titration was performed on the electrolyte as shown in Figure 5.

Figure 5 shows the results of the NO_2^- titration performed on the $NaNO_3$ electrolyte as a function of time during Mg OCV and Mg dissolution (discharge). The Mg electrode rested at OCV for the first 60 minutes of the experiment, followed by a 3.2 mA/cm^2 Mg oxidation (discharge). Aliquots of the $NaNO_3$ electrolyte were removed at 15 minute intervals and titrated to measure the concentration of NO_2^- that was formed. Figure 5 shows that the concentration of NO_2^- in the electrolyte increased with time during both OCV and Mg dissolution, indicating the nitrate to nitrite corrosion reaction (Equation 7) was indeed occurring over the course of the experiment. Interestingly, the rate of NO_2^- production (0.15 μmol cm^{-2} min^{-1}) in the electrolyte was almost twice as large as the rate of H_2 evolution (0.08 μmol cm^{-2} min.$^{-1}$) in the $NaNO_3$ electrolyte from Table I. Hence, the dominant corrosion reaction in the $NaNO_3$ electrolyte was NO_3^- to NO_2^- (Equation 7) instead of H_2 evolution (Equation 5), although some H_2 evolution still occurs. Nevertheless, the total corrosion rate of both reactions in $NaNO_3$ is smaller compared to the overall corrosion rate in the HNa_2PO_4 and NaCl electrolytes, hence $NaNO_3$ is still a relatively favorable electrolyte for Mg-O_2 batteries. The Mg^{2+} faradaic efficiency drops from 93% to 77% when including nitrite corrosion reaction with the H_2 corrosion reaction, which is still more efficient than the other salts studied.

To test Mg-O_2 battery performance in NaCl and $NaNO_3$ electrolytes, a 2-electrode battery cell with an O_2 headspace was constructed and tested. Figure 6a shows the Mg-O_2 battery cell voltage as a function of time during a 5 mA/cm^2 discharge in NaCl and $NaNO_3$ electrolytes. The battery employed a Mg anode, a Whatman filter paper separator, a commercial MnO_2 air cathode (Electric Fuel Ltd.), and a fixed volume of O_2 headspace above the cathode. The OCV of the cell in NaCl and $NaNO_3$ electrolytes was 1.65 V and 1.45 V, respectively, illustrating the lower overall battery cell voltage in the $NaNO_3$ electrolyte compared to NaCl. Additionally, the voltage plateau during discharge was lower in the $NaNO_3$ electrolyte compared to the NaCl electrolyte. However, the $NaNO_3$ discharged to a capacity of ~10 mAh before failure compared to ~6 mAh for the NaCl electrolyte, presumably caused by the lower corrosion rate in $NaNO_3$ compared to NaCl. In Figure 6b, the cumulative gas in the pure O_2 headspace

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is plotted versus time for the same battery cells as in Figure 6a. At OCV, the quantity of gas in the headspace increased with time in the NaCl electrolyte (H_2 evolution from Mg corrosion) and decreased in the $NaNO_3$ electrolyte. During Mg discharge, the gas pressure in the headspace of the cell with the NaCl electrolyte continued to increase as more H_2 was evolved (Mg corrosion) than O_2 was consumed by the MnO_2 cathode. In contrast, during Mg discharge in a $NaNO_3$ electrolyte, the gas pressure in the headspace of the cell decreased with time as O_2 was consumed by the MnO_2 cathode. These results clearly indicate that choice of electrolyte salt in an aqueous Mg-O_2 battery can have a substantial effect on the discharge capacity of the cell. Of note, as a result of the different H_2 evolution rates at OCV and during discharge, it was difficult to precisely calculate the O_2 consumption during battery discharge. Furthermore, H_2 evolution clearly changes as the discharge proceeds due to the change in electrolyte pH and Mg surface film formation during cell operation.

In conclusion, the electrolyte salt choice in Mg-O_2 batteries significantly affects the electrochemistry and gas evolution/consumption during discharge and OCV. A novel electrochemical H-cell with mass spectroscopy and pressure rise capabilities was used to study the fundamental corrosion of Mg in various aqueous electrolytes. HNa_2PO_4 and $NaNO_3$ electrolytes reduced the H_2 evolution corrosion reaction relative to a saline electrolyte, with $NaNO_3$ maximizing Mg^{2+} dissolution from the productive faradaic reaction appropriate for Mg- O_2 batteries at significant current densities. Unfortunately, the $NaNO_3$ electrolyte also displayed a lower OCV voltage and introduced a NO_3^- to NO_2^- corrosion reaction to the battery, as verified by NO_2^- titrations. However, even when accounting for both the NO_2^- and H_2 corrosion reactions, the $NaNO_3$ electrolytes still exhibit faradaic Mg^{2+} efficiencies more than 20% higher than NaCl and HNa_2PO_4. The formation of passivating surface films formed on Mg play a crucial role in corrosion and in Mg^{2+} dissolution during battery discharge and an ideal electrolyte for Mg-O_2 batteries would inhibit corrosion without inhibiting Mg^{2+} dissolution. A potentially interesting direction of future research is the use of either mixed aqueous/nonaqueous or protic/aprotic electrolytes, where it may be possible to more carefully optimize the tradeoff in corrosion inhibition and anodic Mg^{2+} dissolution.

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